EE6309: RENEWABLE ENERGY SYSTEMS

PRE-LAB EXERCISE LABORATORY 03

NAME	: WIJERATHNA S.S.M.
REG. NO	: EG/2022/5421
GROUP NO	: RE 11
DATE	: 15/07/2026

CamScanner

DATE : 15/07/2026

EXPERIMENT : SOLAR THERMAL SYSTEM

01. Describe three ways in which solar energy can be harnessed with modern technology applications.

I. Photovoltaic systems

• Convert sunlight directly into electricity using photovoltaic cell and solar panels produce DC electricity, which is converted to AC.

II. Solar thermal Energy

• Convert solar into heat. Solar collectors absorb sunlight and heat water or another fluid.

III. Passive solar energy.

Use building design to capture and use solar heat naturally. No pumps or electrical equipment are required.

02. Explain the operation of active and passive solar water heating systems.

I. Active solar heating - An active solar heating system uses solar collectors to heat water or a heat transfer fluid.

II. Passive solar water heating system - It heats water using solar collectors and relies on natural thermosiphon effect to circulate hot water to a storage tank without pumps or electricity.

CamScanner